



Marwadi
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FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Engineering

OPEN SOURCE TECHNOLOGIES (01CE0618)

OPEN SOURCE TECHNOLOGIES (01CE0618) Lab Manual

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Batch: B

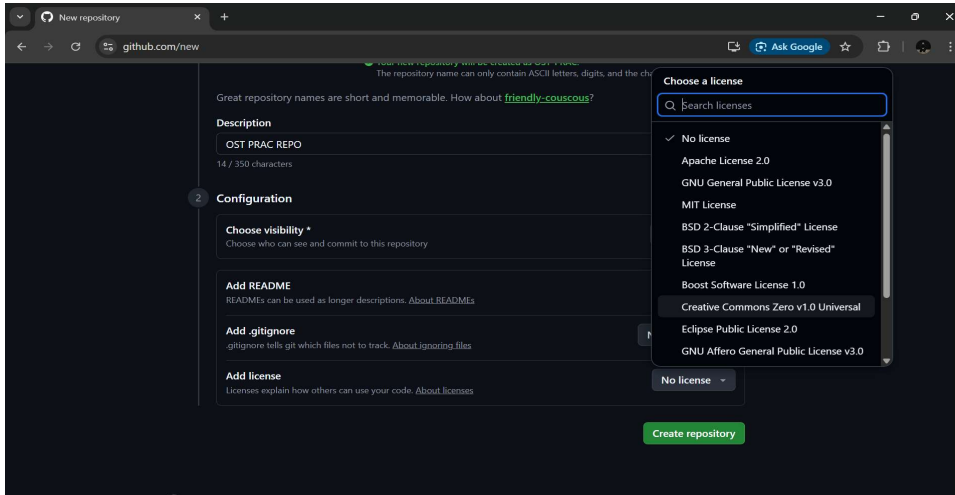
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Experiment 1

AIM: Explore GitHub/GitLab for open-source projects with different licenses

1. List of GitHub Licenses



2. Licenses and Short Description Table

License	Type	Short Description	Best Used For
No License	Proprietary	No one can legally use, modify, or distribute your code	Private or restricted projects
MIT License	Permissive	Free use, modification, distribution (even commercial) with attribution	Student projects, open-source libraries
Apache License 2.0	Permissive	Like MIT + patent protection	Professional & enterprise projects
GNU GPL v3.0	Strong Copyleft	Modified code must remain open-source	Enforcing open-source sharing
GNU GPL v2.0	Strong Copyleft	Same as GPLv3 but without patent rules	Legacy open-source projects
GNU LGPL v2.1	Weak Copyleft	Can be used in closed-source apps; library changes stay open	Libraries & frameworks
GNU AGPL v3.0	Very Strong Copyleft	Server/SaaS usage must publish source code	Web & SaaS open-source apps
BSD 2-Clause	Permissive	Similar to MIT, minimal	Academic &

License	Type	Short Description	Best Used For
License		restrictions	research projects
BSD 3-Clause License	Permissive	BSD 2-Clause + no endorsement clause	Open-source with branding safety
Mozilla Public License 2.0	Weak Copyleft	Only modified files must be open-source	Mixed open/closed projects
Eclipse Public License 2.0	Weak Copyleft	Business-friendly, file-level copyleft	Enterprise & Java projects
Boost Software License 1.0	Very Permissive	Almost no restrictions at all	C++ and utility libraries
Creative Commons Zero (CC0) 1.0	Public Domain	Waives all rights completely	Data, examples, templates
The Unlicense	Public Domain	Places code in public domain	"Do anything" software

3. Licenses Comparison Table

License	Commercial Use	Modify	Distribute	Must Open Source Changes	Patent Protection	Best For
No License	✗	✗	✗	✗	✗	Private / restricted code
MIT	✓	✓	✓	✗	✗	Student & open-source projects
Apache 2.0	✓	✓	✓	✗	✓	Enterprise & professional OSS
GPL v3.0	✓	✓	✓	✓ (entire project)	✓	Strict open-source enforcement
GPL v2.0	✓	✓	✓	✓ (entire project)	✗	Legacy GPL projects
LGPL v2.1	✓	✓	✓	⚠ (library only)	✗	Libraries
AGPL v3.0	✓	✓	✓	✓ (including SaaS)	✓	Web/SaaS open-source apps
BSD 2-Clause	✓	✓	✓	✗	✗	Academic/research code
BSD 3-Clause	✓	✓	✓	✗	✗	Open-source with name protection

License	Commercial Use	Modify	Distribute	Must Open Source Changes	Patent Protection	Best For
MPL 2.0	✓	✓	✓	Ⓜ (modified files only)	✓	Mixed open/closed projects
EPL 2.0	✓	✓	✓	Ⓜ (file-level)	✓	Enterprise / Java projects
Boost 1.0	✓	✓	✓	✗	✗	Utility & C++ libraries
CC0 1.0	✓	✓	✓	✗	✗	Data, examples (not ideal for code)
Unlicense	✓	✓	✓	✗	✗	Public-domain software

4. List of GitHub Alternatives

- GitLab
- Bitbucket
- Gitea
- GitLab Community Edition (CE)
- Codeberg
- SourceForge
- Azure DevOps
- AWS CodeCommit

5. GitHub vs GitLab Table

Feature	GitHub	GitLab
Owner	Microsoft	GitLab Inc.
Type	Code hosting platform	DevOps platform
Repository Hosting	Yes	Yes
CI/CD	✗ (via GitHub Actions)	✓ Built-in
Private Repositories	✓ (free)	✓ (free)
Issue Tracking	✓	✓
Project Management	Basic	Advanced
Self-Hosting	✗	✓
Open-Source Focus	Very high	Very high
Learning Curve	Easy	Moderate
Best For	Students & OSS	DevOps & enterprises

6. Open-Source vs Proprietary vs Freeware

Feature	OSS (Open Source Software)	Freeware	Proprietary Software
Source Code	Available	Not available	Not available
Cost	Free	Free	Usually paid
Modification	Allowed	Not allowed	Not allowed
Redistribution	Allowed	Limited	Restricted
Ownership	Community / Public	Company	Company
Customization	High	Very limited	Very limited
License Type	Open-source licenses	Free-use license	Commercial license
Examples	Linux, Git, Firefox	VLC, Skype	Windows, MS Office

Experiment 02

AIM: Setup Git and explore commands related to Version Control System (VCS)

Code:

```
gadhv@pushpraj MINGW64 ~ (main)
```

```
$ git --version
```

```
git version 2.50.1.windows.1
```

```
gadhv@pushpraj MINGW64 ~ (main)
```

```
$ git init
```

```
Reinitialized existing Git repository in C:/Users/gadhv/.git/
```

```
gadhv@pushpraj MINGW64 ~ (main)
```

```
$ mkdir ost
```

```
mkdir: cannot create directory 'ost': File exists
```

```
gadhv@pushpraj MINGW64 ~ (main)
```

```
$ cd ost
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git init
```

```
Initialized empty Git repository in C:/Users/gadhv/ost/.git/
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ git status
```

```
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
sample.txt
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ echo sample.txt
```

```
sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ echo demo.py
```

```
demo.py
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ git status
```

On branch master

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

sample.txt

nothing added to commit but untracked files present (use "git add" to track)

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ dir
```

```
sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ echo sample.txt
```

```
sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ echo Hello>sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ dir
```

```
sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ echo Hello > sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ ls
```

```
sample.txt
```

```
gadhv@pushpraj MINGW64 ~/ost (master)
```

```
$ cat sample.txt
```

```
Hello
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ cat sample.txt
```

```
Hello
```

```
hello from dev
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git remote add origin https://github.com/Pushprajgadhvi/OST-PRAC
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git remote -v
```

```
origin https://github.com/Pushprajgadhvi/OST-PRAC (fetch)
```



```
origin https://github.com/Pushprajgadhvi/OST-PRAC (push)
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git remote add origin http://github.com/Pushprajgadhvi/OST-PRAC
error: remote origin already exists.
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git remote -v
```

```
origin https://github.com/Pushprajgadhvi/OST-PRAC (fetch)
```

```
origin https://github.com/Pushprajgadhvi/OST-PRAC (push)
```

```
gadhv@pushpraj MINGW64 ~/ost (main)
```

```
$ git push -u origin main
```

```
Enumerating objects: 9, done.
```

```
Counting objects: 100% (9/9), done.
```

```
Delta compression using up to 12 threads
```

```
Compressing objects: 100% (4/4), done.
```

```
Writing objects: 100% (8/8), 768 bytes | 153.00 KiB/s, done.
```

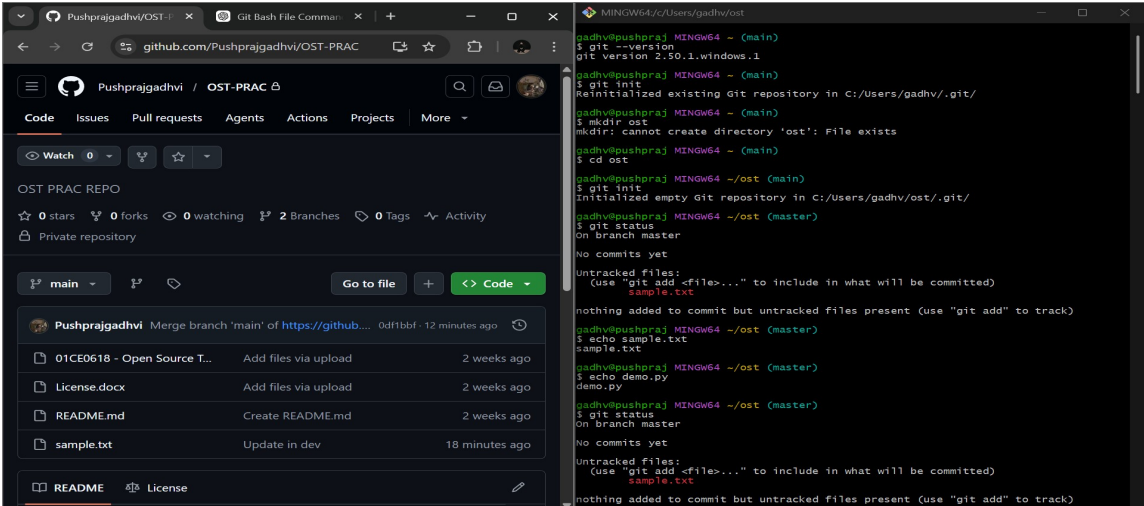
```
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
```

```
To https://github.com/Pushprajgadhvi/OST-PRAC.git
```

```
ff41378..0df1bbf main -> main
```

```
branch 'main' set up to track 'origin/main'.
```



```
gadhv@pushpraj MINGW64 ~ (main)
$ git --version
git version 2.50.1.windows.1

gadhv@pushpraj MINGW64 ~ (main)
$ git init
Reinitialized existing Git repository in C:/Users/gadhv/.git/

gadhv@pushpraj MINGW64 ~ (main)
$ mkdir ost
mkdir: cannot create directory 'ost': File exists

gadhv@pushpraj MINGW64 ~ (main)
$ cd ost

gadhv@pushpraj MINGW64 ~/ost (main)
$ git init
Initialized empty Git repository in C:/Users/gadhv/ost/.git/

gadhv@pushpraj MINGW64 ~/ost (master)
$ git status
On branch master
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  sample.txt

nothing added to commit but untracked files present (use "git add" to track)

gadhv@pushpraj MINGW64 ~/ost (master)
$ echo sample.txt
sample.txt

gadhv@pushpraj MINGW64 ~/ost (master)
$ echo demo.py
demo.py

gadhv@pushpraj MINGW64 ~/ost (master)
$ git status
On branch master
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  sample.txt

nothing added to commit but untracked files present (use "git add" to track)
```